

Yurina Nakazato

Curriculum Vitae (last updated October 2025)

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Research Interests

Galaxy formation and evolution, cosmology, first stars and galaxies, cosmological simulations, multi-wavelength observations

Work Experience

Oct. 2025 - present **Flatiron Research Fellow, Center for Computational Astrophysics**, Flatiron Institute, New York, US

Education

Apr. 2023 - Sep. 2025 Ph.D. student, The University of Tokyo, Japan
Thesis: Formation and Evolution of High-Redshift Galaxies Using Cosmological Simulations and Multi-Wavelength Observations
Advisor: Naoki Yoshida

Apr. 2021 - Mar. 2023 M.S. in Physics, The University of Tokyo, Japan
Thesis: *Formation and evolution of star clusters and galaxies in the early Universe*
Advisor: Naoki Yoshida
President's Award (1/3000), Dean's Award (1/128)

Apr. 2017 - Mar. 2021 B.S. in Physics, The University of Tokyo, Japan

Fellowships and Awards

Oct. 2025 The L'Oréal-UNESCO For Women in Science Japan Fellowship

Aug. 2025 The Excellence Award for Final Exam

Jan. 2025 Flatiron Research Fellow

Jan. 2025 CITA Fellow (declined)

Feb. 2024 Outstanding Poster Award at the International Symposium on Quantum Electronics 2024

Apr. 2023- present JSR fellowship

Apr. 2023- present Japan Society for Promotion of Science (JSPS) Research Fellow, DC1

Mar. 2023 The School of Science Encouragement Award

Mar. 2023 The University of Tokyo President's Award

Mar. 2023 The Excellence Award for Qualifying Exam

Oct. 2021- present The International Graduate Program for Excellence in Earth-Space Science (IGPEES), the University of Tokyo

Aug. 2021 Best Oral Presentation Award at 51th astronomical meeting for junior researchers

Leadership & Outreach

Jun. 2023 Organizer of Get-Together Event for Women in STEM, the University of Tokyo

Jul. 2022 Public Talk at Women in STEM, School of Science, the University of Tokyo

Dec. 2019 Rikejo Initiative, the University of Tokyo

Oct. 2019 Invited talk at Dow Chemical Company, Tokyo, Japan

Grants

Apr. 2023-Mar.2026	Evolution of first galaxies with multi-wavelength observations and numerical simulations, 4.2M JPY (30K USD), JSPS Grant-in-Aid for Early-Career Scientists, No. 23KJ0728
Apr. 2023-Mar.2026	Grand Aid from JSR fellowship, 3M JPY (21K USD)

Talks at Conferences and Workshops (Selected)

total: 25 (24 oral talks, 9 peer-reviewed talks)

Sep. 2024	ILR workshop 2024 @Osaka: From Galaxy Formation to Supermassive Black Holes , Osaka, Japan
Aug. 2024	IAUS391 , Cape Town, South Africa
Jul. 2024	European Astronomical Society (EAS) Annual meeting 2024 , Padova, Italy
Feb. 2024	The International Symposium on Quantum Electronics 2024 , poster, The University of Tokyo, Japan
Nov. 2023	Resolving the Extragalactic Universe with ALMA & JWST , Waseda Univ., Japan
Jul. 2023	Shedding new light on the first billion years of the Universe , Marseille, France
Sep. 2022	The 9th East Asian Numerical Astrophysics Meeting (EANAM9) , Okinawa, Japan
Aug. 2022	Star Formation in Different Environments (SFDE) 2022 , Quy Nhon, Vietnam
Jul. 2022	National Astronomy Meeting (NAM) 2022 , online, hosted by the University of Warwick
Nov. 2021	IAU Symposium 362: The predictive power of computational astrophysics as a discovery tool , online
Aug. 2021	RESCEU Summer School 2021 , online, hosted by the University of Tokyo

Seminars (Selected)

Dec. 2024	One-day workshop on high-z cosmological simulations , The University of Tokyo
Jul. 2024	Special seminar , JSR corporation, Online
Apr. 2024	Special seminar , Scuola Normale Superiore, Italy
Dec. 2024	Special seminar , Universidad Autónoma de Madrid , Spain
Dec. 2024	Special seminar , Centro de Astrobiología, (CAB, CSIC-INTA), Spain
Aug. 2023	One-day workshop on galaxies in the era of JWST/ALMA , The University of Tokyo
Mar. 2023	Supersonic Project: Collaboration meeting , UCLA, US
Feb. 2023	Special seminar , UCLA, US

Coverage in Media

Hunting Faint Galaxies with JWST as a Test for the Dark Matter Model based on Williams et al. 2024
- *Bright galaxies put dark matter to the test*, UCLA

The core of the most distant galaxy protocluster observed by JWST & ALMA based on Hashimoto et al. 2023
- *The Strong Tag Team of the James Webb Space Telescope and ALMA Captures the Core of the Most Distant Galaxy Protocluster*, Kavli IPMU, Tsukuba Univ., Waseda Univ., Nagoya Univ.

Teaching & Professional Service

Teaching

Apr. 2021-Aug. 2021	Teaching Assistant of Fluid Mechanics, the University of Tokyo
Apr. 2019- Jan. 2024	Language Assistant, International Lounge, the University of Tokyo

Professional Service

Apr. 2025	Organizer, One-day workshop on for stellar-cluster simulation, The University of Tokyo, Tokyo, Japan
Dec. 2024	Organizer, One-day workshop on high-z cosmological simulations, The University of Tokyo, Tokyo, Japan
Aug. 2023	Organizer, The 53rd Summer School on Astronomy and Astrophysics, The University of Tokyo, Tokyo, Japan
Mar. 2023	Workshop Organizer, Astrophysics Workshop for Young Researchers, The University of Tokyo, Tokyo, Japan

Proposals

co-Investigator on ALMA proposals

4. 2023.1.00843.S, PI: Y.Tamura, 15.6h
Disentangling the ionization structure of HII regions in a density-bounded Ly α emitter in the Reionization Era
3. 2023.1.00193.S, PI: Y. Fudamoto, 17.3h
[CII] 158 μ m emission line and dust observation of the most distant known overdensity of galaxies at $z = 7.88$
2. 2023.1.00235.S, PI: Y. Fudamoto, 10.0 h
The ALMA-JWST synergy: [OIII]88 μ m and [CII]158 μ m emission line observations of $z = 9.51$ galaxy identified by JWST
1. 2023.1.01362.S, PI: Y. Tamura, 25.9 h
Pandora's ELPIS: The Emission-Line Protocluster Imaging Survey of the furthest overdensity beyond Pandora's Cluster

Research & industry experience

Apr. 2024 - Jul. 2024	Visiting Student at Scuola Normale Superiore di Pisa - Host researcher is Prof. Andrea Ferrara.
Nov. 2023 - Dec. 2023	Visiting Student at Universidad Autónoma de Madrid - Host researcher is Prof. Daniel Ceverino. - Worked on formation and evolution of clumpy galaxies by using FirstLight simulation
Feb. 2023 - Mar. 2023	Visiting Student at UCLA - Host researcher is Prof. Smadar Naoz. - Worked on Supersonic Project [link]
Jun. 2020 - Mar. 2021	Study and Visit Abroad Program - Funded by the faculty of science, the University of Tokyo - Online research internship at Naoz lab, UCLA
Dec. 2019 - Jan. 2020	Online Language Exchange Program - Utokyo & TUM- - Participated in an online international exchange program with students from Technical University of Munich
Jun. 2019 - Sep. 2019	UTokyo Global Internship Program

- Funded by DAIKIN, a company leading air conditioning and refrigeration.
 - Two-month workplace training at DAIKIN Japan and two-week research internship at DAIKIN Europe in Belgium. Business proposal for food loss and integrated solution of air conditioning and refrigeration.
- Aug. 2019 **Summer School of Particle Physics and Nuclear Physics**
- Funded by the High Energy Accelerator Research Organization.
 - Experiment of measuring muon decay time and observing Lamor Precession and Single-Spin Asymmetries. Made the final presentation and poster session.
- Aug. 2019 **Nanotechnology Platform Student Training Program**
- Funded by National Institute for Materials Science.
 - Five-day research program at Spring-8, the world's largest third-generation synchrotron radiation facility. Conducted X-ray photoelectron spectroscopy (XPS) experiments and data analysis. Made a presentation at the University of Tokyo in September.
- Feb. 2019- Mar. 2019 **Undergraduate Research Assistant, TOMODACHI STEM Program**
- Funded by U.S.-Japan Council.
 - Five-week science & engineering research internship at Gerts lab, Rice University, Houston. Researched and analyzed the heavy iron collision data of STAR experiment conducted at BNL.
 - Final week study tour to Washington, DC including site visits to the Society for the Promotion of Science, JAXA, U.S.-Japan Council, and Women in STEM Workshop at Lehigh University.
- Jul. 2014- Aug. 2014 **Okinawa Global Leaders Program**
- Funded by Okinawa Prefectural Board of Education.
 - Three-week program to introduce students to key concepts in intercultural communication and global leadership.

--- Active collaborations

- Supersonic Project
- FirstLight simulations
- Reionization and the ISM/Stellar Origins with JWST and ALMA (RIOJA, JWST GO1840)

--- Competencies

Scientific programming : Python, C++, Fortran, Latex
 Simulation software : SKIRT, AREPO, ART
 Astrophysics tool : CLOUDY, SKIRT, Astrodendro, A-SLOTH

--- References

Prof. Naoki Yoshida (current supervisor), The University of Tokyo, naoki.yoshida@ipmu.jp
 Prof. Andrea Ferrara, Scuola Normale Superiore, andrea.ferrara@sns.it
 Prof. Smadar Naoz, University of California, Los Angeles, snaoz@astro.ucla.edu